

MUHAMMAD IKRAM

Software Engineer | Full-Stack | Applied AI | Automation

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Portfolio: ikram-neon.vercel.app

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PROFESSIONAL SUMMARY

Software Engineer building production-oriented web, AI, and automation systems across SaaS, healthcare, POS, e-commerce, and computer vision. Delivered a portfolio of **35+ end-to-end projects**, including **6 live deployments**, with hands-on ownership from data modeling and API design to frontend delivery and cloud deployment. Strong in Python, TypeScript, Next.js, FastAPI, and PostgreSQL, with applied AI experience in Retrieval-Augmented Generation (RAG), tool-using agents, and real-time vision systems.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, PHP, Dart, C++, SQL, HTML5, CSS3
AI & ML	RAG, Agentic AI, ReAct, tool execution, Groq, Google Gemini, Ollama, Pinecone, ChromaDB, YOLOv8, ONNX, OpenCV, LangChain, Hugging Face, PyTorch, TensorFlow
Backend & APIs	FastAPI, Next.js API Routes, Node.js, Laravel, Django, REST APIs, WebSockets, Webhooks, Docker
Frontend	Next.js 16, React 19, Tailwind CSS, shadcn/ui, Zustand, Vite, Inertia.js, Streamlit, React Three Fiber
Databases & Data	PostgreSQL, MySQL, SQLite, Redis, Prisma ORM, JSON-based persistence
Mobile, Automation & DevOps	Flutter, Playwright, browser automation, Git, GitHub Actions, CI/CD, Vercel

PROFESSIONAL EXPERIENCE

Independent Software & AI Engineer | Self-Directed Project Portfolio 2023 – Present | Swat, Pakistan

- Architected, built, and deployed **35+ software projects** spanning applied AI, SaaS, enterprise systems, mobile apps, automation, and data-driven web platforms.
- Maintained **6 live deployments** and handled end-to-end delivery across architecture, databases, APIs, frontend engineering, deployment, testing, and iteration.

Full-Stack WordPress Developer | Freelance Jun 2024 – Aug 2024 | Remote

- Delivered responsive WordPress and WooCommerce websites aligned with client requirements, usability, and mobile performance.
- Improved backend performance and technical SEO, configured payment and contact workflows, and shipped production-ready deployments.

FEATURED ENGINEERING PROJECTS

Agentic ChatBot – University of Swat [Live demo](#)

[Python](#), [FastAPI](#), [React](#), [Groq](#), [Pinecone](#), [RAG](#)

Flagship KPITB Generative AI Fellowship Project

- Built a grounded campus assistant for admissions, eligibility, fees, and university information with source-backed answers and document citations.
- Combined Pinecone retrieval with live web context and Groq inference; added similarity thresholds and refusal behavior to reduce unsupported responses.

BiteBox POS – Restaurant Management System [Live demo](#)

[Next.js 16](#), [React 19](#), [TypeScript](#), [Zustand](#)

Production fast-food POS and kitchen workflow

- Delivered touchscreen ordering, cash/card/mobile-money checkout, change calculation, kitchen Kanban flow, discounts, loyalty, and operational analytics.
- Implemented recipe-based inventory deduction plus A4 and 80mm thermal receipts with lightweight JSON persistence for simple deployment.

Trade Automation & SaaS Platform

[Live demo](#)

Laravel, PHP, React, Inertia.js, Stripe, Telegram API

Commercial trading-indicator and membership platform

- Engineered subscription, licensing, Stripe verification, TradingView webhook workflows, Telegram community access, and administrative telemetry.
- Built automation around entitlement management and indicator access to reduce manual fulfillment and improve operational visibility.

Hospital Management System

[Live demo](#)

Next.js 16, React 19, TypeScript, Prisma, PostgreSQL

Healthcare administration and electronic records platform

- Built patient intake, doctor scheduling, pharmacy inventory, ward/bed management, billing, and clinical record workflows across five role-based dashboards.
- Added appointment rescheduling, notification workflows, and scoped access controls across doctors, nurses, admins, receptionists, and patients.

LUMIERE – Premium 3D E-Commerce

[Live demo](#)

Next.js 16, React 19, React Three Fiber, Prisma, Stripe

Interactive high-end commerce storefront

- Created an interactive storefront with WebGL hero visuals, 360-degree product presentation, faceted filtering, wishlist sync, and a structured multi-step checkout.
- Supported Stripe, EasyPaisa, JazzCash, and cash-on-delivery flows while maintaining responsive performance across devices.

Nexus Traffic Management System

Python, Django, YOLOv8, ONNX, WebSockets, React

Smart-city real-time surveillance and violation detection

- Developed real-time vehicle violation detection for speeding, phone use, and seatbelt compliance across multi-camera streams.
- Streamed annotated feeds through WebSockets and added emergency-priority signal overrides with hotspot analytics for operational monitoring.

PhoenixAgent – Autonomous Desktop AI Assistant

Python, PyQt6, Groq, Faster-Whisper, ChromaDB

Voice-enabled tool-using desktop agent

- Built ReAct-style orchestration over 50+ operating-system, browser, and filesystem tools with safety checks around higher-impact actions.
- Integrated local speech recognition, text-to-speech, OCR screen awareness, vector memory, and multi-provider fallback across Gemini, Groq, and Ollama.

ADDITIONAL SELECTED SYSTEMS

ERP backend with accounting, inventory, and tenant RBAC • OEC agentic automation and RAG chatbot • Khanaqa Seminary CMS • LinkedIn auto-apply agent • Dynamic-site browser automation engine • TailorTrack Flutter app • Food ordering platform • Payroll system

EDUCATION

Bachelor of Science in Software Engineering (BS-SE)

Nov 2024 – Present

University of Swat, Pakistan

Higher Secondary School Certificate (FSc, Computer Science)

2022 – 2024

Government Degree College, Mingora, Swat, Pakistan

CERTIFICATIONS & FELLOWSHIP

- **KPITB Generative AI Fellowship** – Khyber Pakhtunkhwa Information Technology Board (Deep Learning, NLP, Transformers, RAG, Agentic AI)
- **Artificial Intelligence Certification** – National Vocational and Technical Training Commission (NAVTTTC), Distinction, 89/100
- **Python for Data Science** – Great Learning Academy